

MOHAMMED ECH-CHAIBI

Junior Automation & Mechatronics Engineer

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PROFESSIONAL SUMMARY

Junior Automation & Mechatronics Engineer with skills in PLC programming, industrial robotics, embedded systems and computer vision. Experienced in designing automation sequences, developping HMI/PLC interfaces and producing engineering documentation (GAMP V). Strong foundation in control logic, system integration and industrial process optimization. Motivated to contribute to smart manufacturing and advanced automation projects.

WORK HISTORY

Automation engineer, University of Limerick	12/2025 – Present Limerick
- Designed an autonomous robotic harvesting system integrating a Mitsubishi 6-axis robot, machine vision, and hydroponic monitoring. - Developed OpenCV-based detection and alignment to identify harvestable plants and improve robot positioning. - Integrated the GuardLogix control and safety design, including interlocks, emergency stop logic, and operator feedback.	
Automation engineer, University of Limerick	10/2025 – 12/2025 Limerick
- Programmed a Studio 5000 PLC to control a multi-step handling sequence for cube orientation, rotation, and inspection. - Designed HMI screens for operation, status monitoring, alarms, and manual control, following GAMP 5 good practice principles. - Implemented safety functions including E-stop, pause/resume logic, interlocks, and alarm handling in line with IEC 61508 concepts.	
Automation engineer, University of Limerick	02/2026 – 03/2026 Limerick
- Designed and implemented a conveyor-based bottle sorting system to classify bottles by orientation and reject knocked-down units automatically. - Integrated through-beam and proximity sensors with a pneumatic cylinder and solenoid valve for real-time detection, position identification, and rejection. - Developed LabVIEW/myRIO control logic and built the full system, including conveyor wiring, sensor integration, HMI bottle counting, and emergency stop safety functionality.	
Intern - manufacturing engineer, Flying Whales	04/2025 – 09/2025 Bordeaux
- Modeled the airship's evolving mass and calculated helium quantities across 50 assembly stages, including min/max helium scenarios, while ensuring tilt, CG, and buoyancy compliance. - Developed original formulas to compute helium height in 14 cells and introduced a new imbalance- detection method to make the inflation process more accurate and easier to operate. - Analyzed how factory temperature and pressure variations affect helium volume and behavior, and initiated a risk assessment to ensure the robustness of the helium filling sequence.	

Intern -Automation and Process Improvement, SITI TEA

05/2024 – 09/2024

- Analyzed production workflows using Lean tools to improve system efficiency and operational flow.
- Developed standardized work instructions and proposed layout and process optimizations to simplify operations and reduce non-value-added activities.
- Performed PID tuning on a dosing system (dispenser) to stabilize flow control, improve dosing accuracy, and ensure repeatable process performance.

Marrakech

EDUCATION

Master of engineering: Mechatronics, UNIVERSITY OF LIMERICK

09/2025 – 08/2026

- Coursework in robotics, automation, control systems, embedded systems, and machine vision, with practical work in industrial robotics, PLCs, sensors, and system integration.

LIMERICK

Engineering Degree: Industrial Engineering, IMT NORD EUROPE

09/2023 – 08/2025

- Focus on industrialisation, manufacturing systems, process optimisation, and 3D modelling, with projects in automation, production flow analysis, and prototyping.

Douai

CPGE: Engineering, Ibn Timiya

09/2021 – 09/2023

- Two-year intensive preparatory programme focused on advanced mathematics, physics, and engineering fundamentals.

Marrakech

SKILLS

Automation & PLC

Allen-Bradley Studio 5000, Siemens TIA Portal, State machines & structured sequence design, Emergency stop logic, safety interlocks, Sensor/actuator control&wiring basics.

Robotics

Mitsubishi RT Toolbox/Melfa Basic, Robot calibration, Robot-PLC communication, 3D Modeling.

Software&Tools

Python, C/C++,sensor integration, MATLAB, Git, Serial TCP communication.

Industrial engineering

Lean principles, process analysis, KPI dashboarding, workflow optimisation.

Soft Skills

Problem-solving, Fast learning & adaptability, Analytical Thinking, Teamwork & coordination.

LANGUAGES

French

Fluent

English

Fluent

Spanish

Elementary

HOBBIES AND INTERESTS

• Chess

• Cooking

• Tennis

CERTIFICATIONS

• Lean Six Sigma Green Belt

• MOOC in Project Management.